AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 32 · TEACHER GUIDE

Human AI Workflow

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Teach oversight as an informed authorised decision.

Supply complete handoffs, preserve stop conditions and assess a fresh action-scope change.

Keep one habit: Ask who saw which evidence and which exact content/version/destination their decision covered.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Explain the **mechanism** and terminology of human review and consequential use

Two 45-minute sessions. Human review needs a responsible actor, **evidence** packet and explicit authority. Adding a person to a **workflow** does not guarantee effective oversight.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can explain human ai **workflow** using a traced example and a stated limitation.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Lesson and workbook; paper and pencil
- Calculator and optional local Python 3 for arithmetic; no live API or private data
- Prepare the fresh retry separately and withhold its solution until after collection

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of untrusted **input** boundaries

Reference example, not the only acceptable wording: Teacher explanation: Untrusted **input** must be treated according to a contract before reaching code, queries or tools.

Data that resembles instructions does not acquire execution authority. For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code.

Reject invalid length or characters.

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of copyright and permitted sources

Reference example, not the only acceptable wording: Teacher explanation: Permission to access a work and permission to redistribute or modify it are separate.

Use a classroom licence ledger instead of assuming all online material is reusable. For this fictional policy, licence L permits classroom display with attribution but forbids public redistribution.

Licence M permits adaptation and redistribution with attribution. Unknown permission means seek approval or choose a permitted alternative; this exercise is not a legal ruling.

Without opening the old lesson, explain one key idea from Evaluation Sets. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of system evaluation sets

Reference example, not the only acceptable wording: Teacher explanation: A system evaluation set contains cases with expected behaviour and scoring rules.

It should cover retrieval, support, fallback and permissions rather than only easy answerable questions. Record case ID, **evidence** version, expected claims or fallback, permitted actions and scoring rubric.

Separate development from final cases. Aggregate metrics only after checking required conditions individually.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Safety Boundaries

Copyright and Provenance

Evaluation Sets

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A student asks a helper to submit an unchecked generated essay under their name. Design a **workflow** that preserves student reasoning and follows classroom rules.

Concept to explain

Allocate tasks according to strengths, **accountability**, and **verification** needs.

Term	Meaning
workflow	An organised sequence of tasks, reviews and handoffs used to complete work.
accountability	Responsibility for explaining decisions, checking outcomes and addressing mistakes.
verification	Checking a claim or result against an appropriate independent source, calculation or observation.
sign-off	Explicit recorded approval that specified work meets the required review conditions.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach responsibility through exact handoff records

> Keep draft authority, validation and final action ownership distinct.

State the owner and stop map

Room-workflow stage	Owner, input and stop rule
Prepare evidence	Learner selects permitted current room sources and states the request
Draft	Assistant proposes content; no approval or submission authority
Validate	Application checks schema, same-room support, access and permitted action; failures stop
Review	Coordinator sees exact version, sources, changes, uncertainties, intended action and destination
Approve	Only coordinator may approve the exact room/content/version/action; no implied approval of changes
Submit	Only coordinator may submit the matching approved version under this toy contract
Revise	Any material content, action or destination change returns to review; no automatic reuse of old approval

Human review needs a responsible actor, **evidence** packet and explicit authority. Adding a person to a **workflow** does not guarantee effective oversight.

For this fictional **workflow** the assistant drafts a room request but only the coordinator can approve and submit it. **Review packet**s include sources, uncertainty, requested action and the exact version.

Any material change requires new approval.

Trace the room approval mismatch

Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval.

The submission violates **version-bound approval**. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Read the H1/H2 handout case

A human approval belongs to the exact draft and action reviewed.

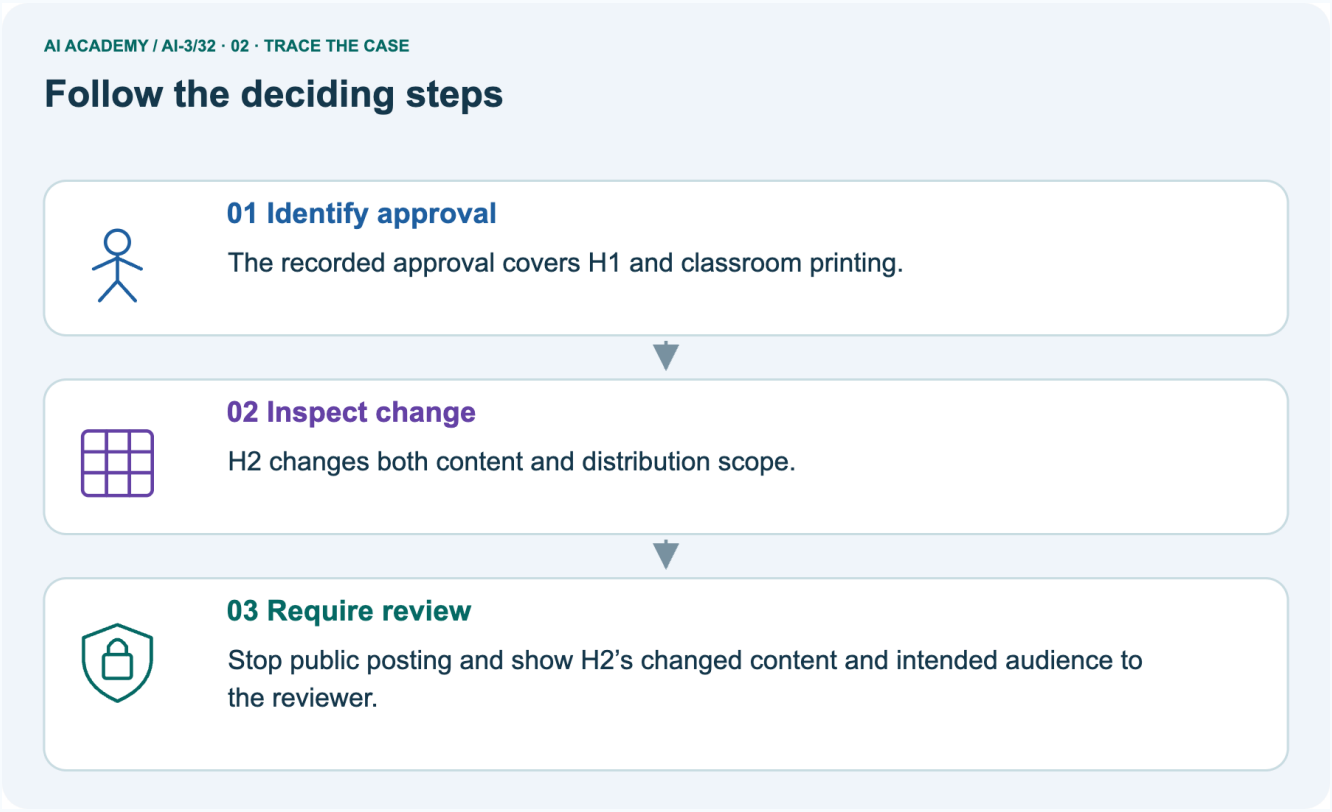
A teacher approves study handout H1 for classroom printing. H2 adds a pupil email address and proposes public posting. No H2 approval exists.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Separate handout fixture: teacher approval covers only H1 and classroom printing. H2 adds a fictional pupil-email placeholder and public posting; no actual address or H2 approval is supplied. Stop the changed action and show the content/destination differences to the authorised reviewer.

Returning to unchanged H1 and classroom printing matches the supplied original scope; no revocation or expiry is stated in this toy case. This does not establish that any printing or posting occurred.

Trace content and distribution changes



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Identify approval	The recorded approval covers H1 and classroom printing.
2. Inspect change	H2 changes both content and distribution scope.
3. Require review	Stop public posting and show H2′s changed content and intended audience to the reviewer.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Stop public posting and show H2′s changed content and intended audience to the reviewer. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict unchanged approved scope

Change: H2 is replaced with unchanged H1 for classroom printing.

Prediction: The original approval covers that declared action.

Reason: Both the approved content and its distribution scope now match.

State effective-review limits

Human review helps only when the reviewer sees the material changes, **evidence** and actual consequences.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Additional discussion · Learner responsibility

An AI drafts, verifies and submits a student's assessment without review.

Where should responsibility remain?

The student reviews, verifies, follows class rules and remains responsible. Submission requires authorisation and must not bypass learning.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. AI drafts questions from approved notes.

2. Learner checks clarity; retrieval **evidence** supports claims.

3. Teacher decides whether material is appropriate.

4. Product records who verified and approved without shifting **accountability** to AI.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Exact explanation to model

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Review packet	Evidence and proposed action supplied to a reviewer
Version-bound approval	Permission tied to an exact proposal version
Escalation	Routing a decision to an authorised responsible person

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Human AI Workflow

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the explanation, then remove help

> Collect each learner’s reasoning before showing feedback.

Main completion prompt

Completion task: Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Expected reasoning and fading help

Teacher check: Teacher explanation: The submission violates **version-bound approval**. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Human review and consequential use. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Why can H1's approval not automatically approve H2? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The reviewer did not approve the added personal detail or public audience. Approval must match the version and action. Changed-case check: The original approval covers that declared action.

Both the approved content and its distribution scope now match.

Changed-case reasoning: The original approval covers that declared action. Both the approved content and its distribution scope now match.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ **Earlier wording, alternatives and picture transcripts**

Demonstrate and guide practice

Board sequence

Write the given facts: An AI drafts, verifies and submits a student's assessment without review.

Write the question to explain: Where should responsibility remain?

Reveal after **prediction**s: The student reviews, verifies, follows class rules and remains responsible. Submission requires authorisation and must not bypass learning.

Use the earlier changed question as optional later practice. Reserve Inspect the new report-distribution scope independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: The submission violates **version-bound approval**. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Qualification: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage	Teacher action
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Room cards A eligible coordinator action, no actual submission; B missing source; C unsupported capacity; D changed version; E wrong actor; F inadequate evidence packet. H1 approval covers unchanged classroom printing only, not H2 public posting. Independent R1 read-only approval does not cover R2 email distribution; reviewer needs exact content/version, recipients, purpose and authorised approval before any send.

A person in the loop is not automatically effective oversight.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the six complete room-**workflow** cards and explicit coordinator-only approval/submission rule. Assign owners, preserve exact content/version/action, show the review packet and trace evidence failures or wrong-actor stops. These are supplied paper simulations, not actual approvals or submissions.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The worked result establishes every future case: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Inspect the new report-distribution scope independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Teacher approves read-only report R1. System adds email distribution in R2. What permission is missing and what should the reviewer see?

Teacher explanation: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Human review needs a responsible actor, **evidence** packet and explicit authority. Adding a person to a **workflow** does not guarantee effective oversight.

For this fictional **workflow** the assistant drafts a room request but only the coordinator can approve and submit it. **Review packet**s include sources, uncertainty, requested action and the exact version.

Any material change requires new approval.

W2 Trace the worked case

Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The submission violates **version-bound approval**. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

W3 Audit the limit

A report omits this qualification: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

The student reviews, verifies, follows class rules and remains responsible. Submission requires authorisation and must not bypass learning.

Additional transfer practice question

A **workflow** drafts a class email, checks its claims and then sends it. Where should approval occur, and what should the approver see?

Reasoned practice answer

Approval must occur before sending. The person should see the exact recipients, subject, message and relevant **verification** results.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Approval of a general idea also approves any later message and recipients.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Student plans and drafts, uses permitted assistance, verifies claims and sources, revises in their own understanding and discloses assistance as required by class rules.

Submission remains an authorised student action; the tool must not bypass review or the learning task.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Teacher approves read-only report R1. System adds email distribution in R2. What permission is missing and what should the reviewer see?

Teacher explanation: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

AI drafts; sources ground; validators check; learner chooses; teacher governs classroom use.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support while preserving the reasoning goal

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending. Require intermediate reasoning and the stated scope.

Fresh retry: Approval permits sending to one named recipient; proposed action adds two more. Expected reasoning: Recipient scope changed; block the expanded send and request review of the new destination list.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Plan the delayed check

Suggested return points in this level: ai-3/33, ai-3/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 33: Capstone Design. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 33: Capstone Design. Carry forward the user need, **evidence** boundaries, owner map and exact review/approval rules into a buildable project plan.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Verified Study Buddy

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Explain the **mechanism** and terminology of human review and consequential use

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Practice and assessment reminders

Before class, solve the supported practice: Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The submission violates **version-bound approval**. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Check every learner before discussion: Use the worked case for Human review and consequential use. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Review effectiveness depends on usable **evidence**, time and authority; a checkbox is not **evidence** of informed review.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: H2 is replaced with unchanged H1 for classroom printing. Require a **prediction** and a reason before comparing. Expected result: The original approval covers that declared action.

Reason: Both the approved content and its distribution scope now match.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

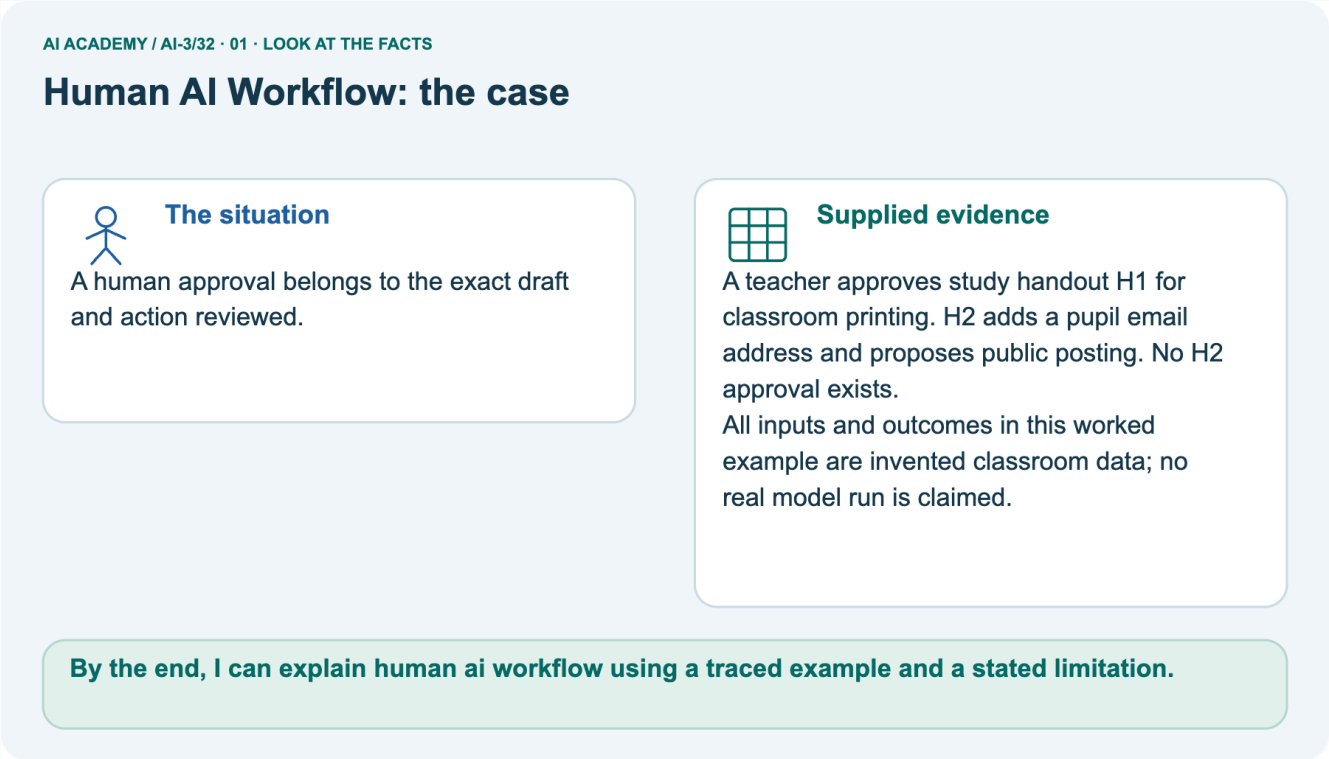
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Human AI Workflow: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

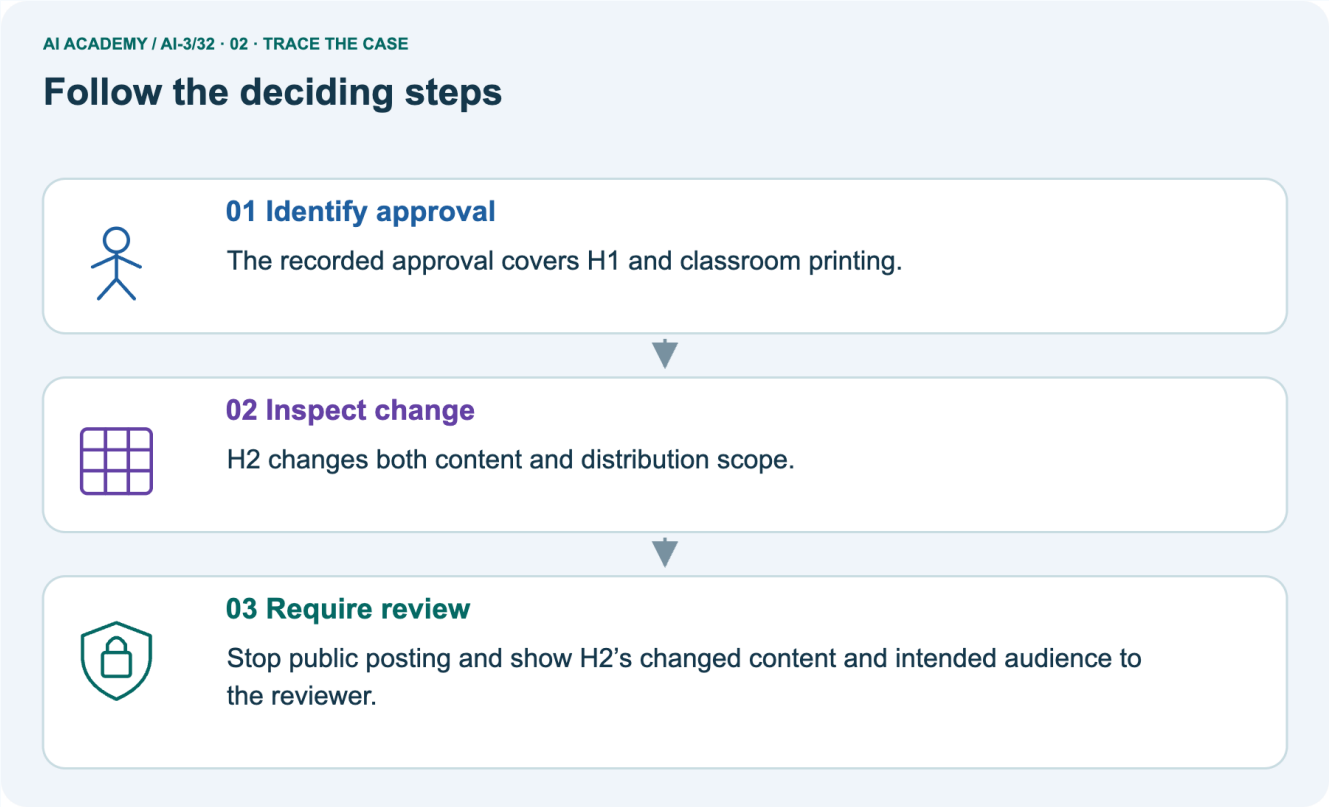
A human approval belongs to the exact draft and action reviewed. Supplied facts: A teacher approves study handout H1 for classroom printing. H2 adds a pupil email address and proposes public posting. No H2 approval exists.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



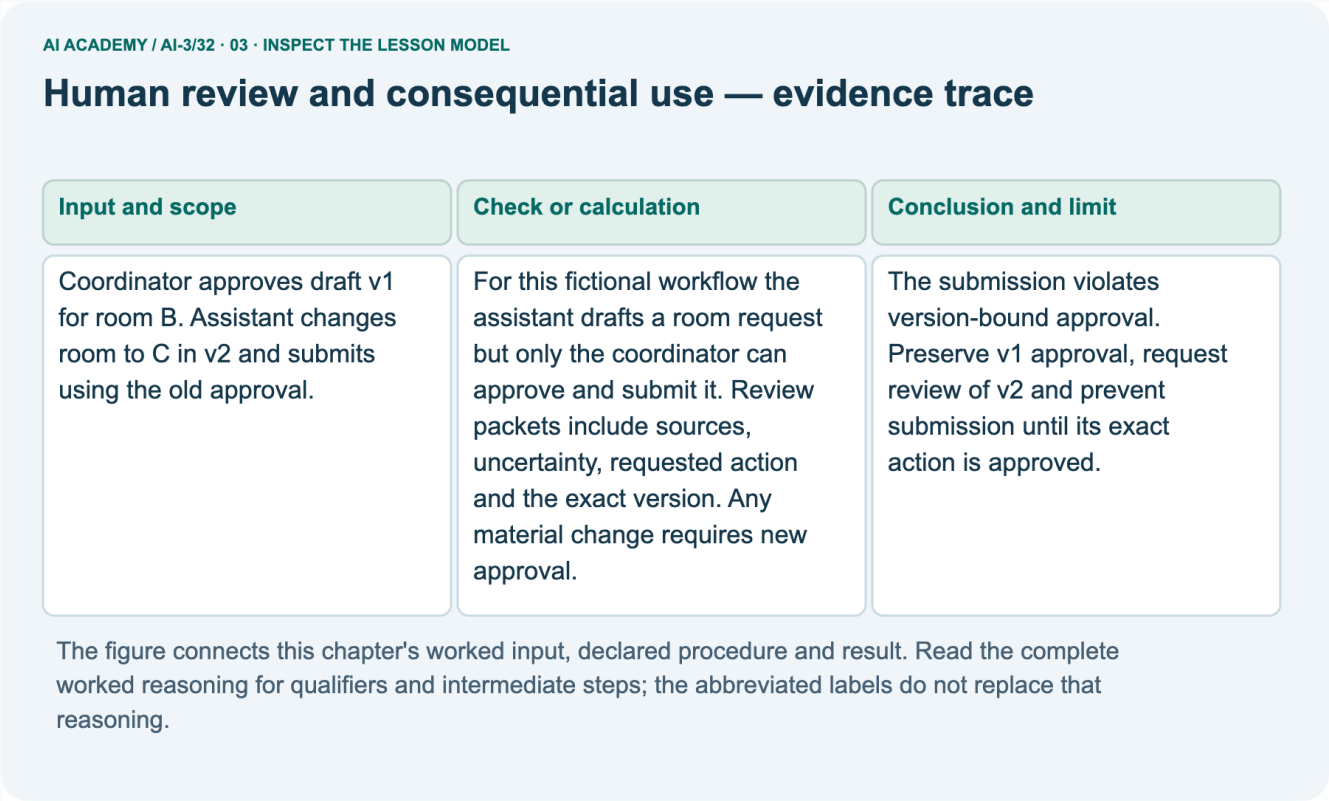
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify approval: The recorded approval covers H1 and classroom printing. → Inspect change: H2 changes both content and distribution scope. → Require review: Stop public posting and show H2′s changed content and intended audience to the reviewer.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Human review and consequential use — evidence trace



AI ACADEMY / AI-3/32 · 03 · INSPECT THE LESSON MODEL

Human review and consequential use — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval.	For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version. Any material change requires new approval.	The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Human review and consequential use — **evidence** trace. Coordinator approves draft v1 for room B.

Assistant changes room to C in v2 and submits using the old approval. | For this fictional **workflow** the assistant drafts a room request but only the coordinator can approve and submit it.

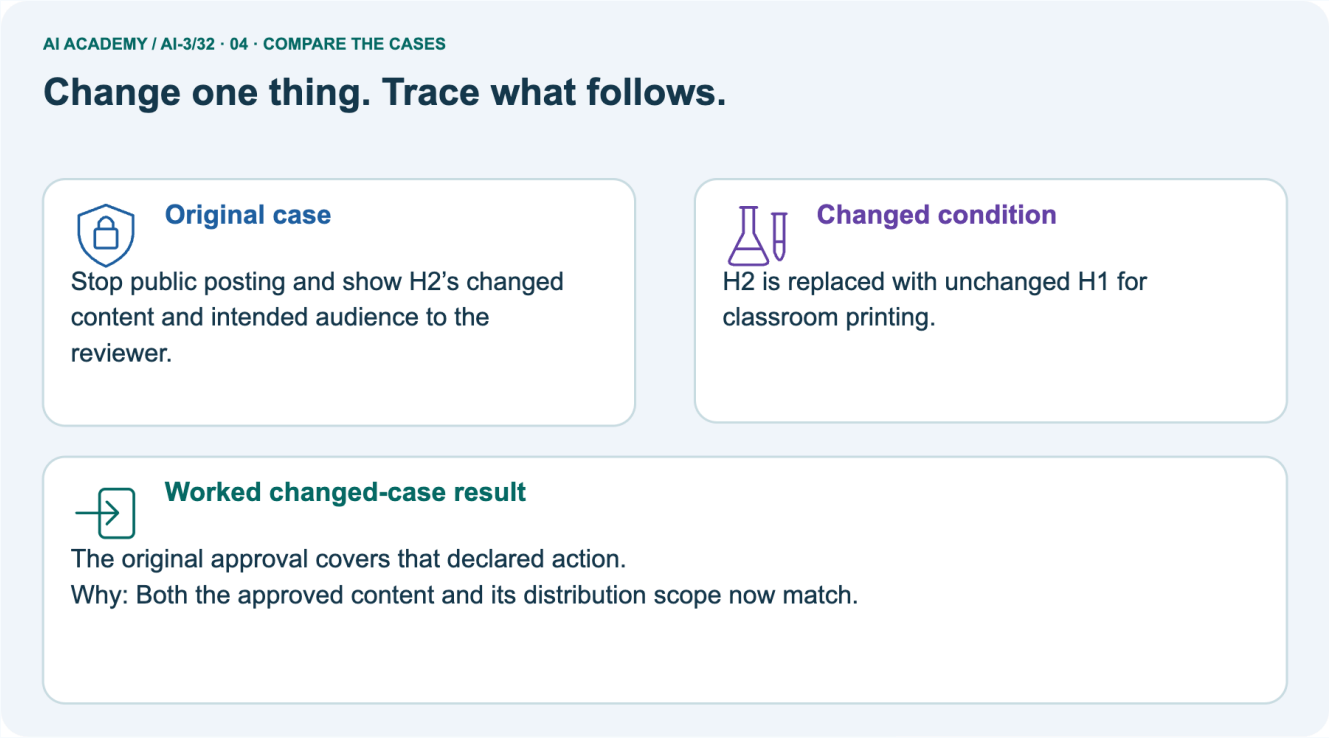
Review packets include sources, uncertainty, requested action and the exact version. Any material change requires new approval. | The submission violates **version-bound approval**.

Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Stop public posting and show H2′s changed content and intended audience to the reviewer. Changed condition: H2 is replaced with unchanged H1 for classroom printing.

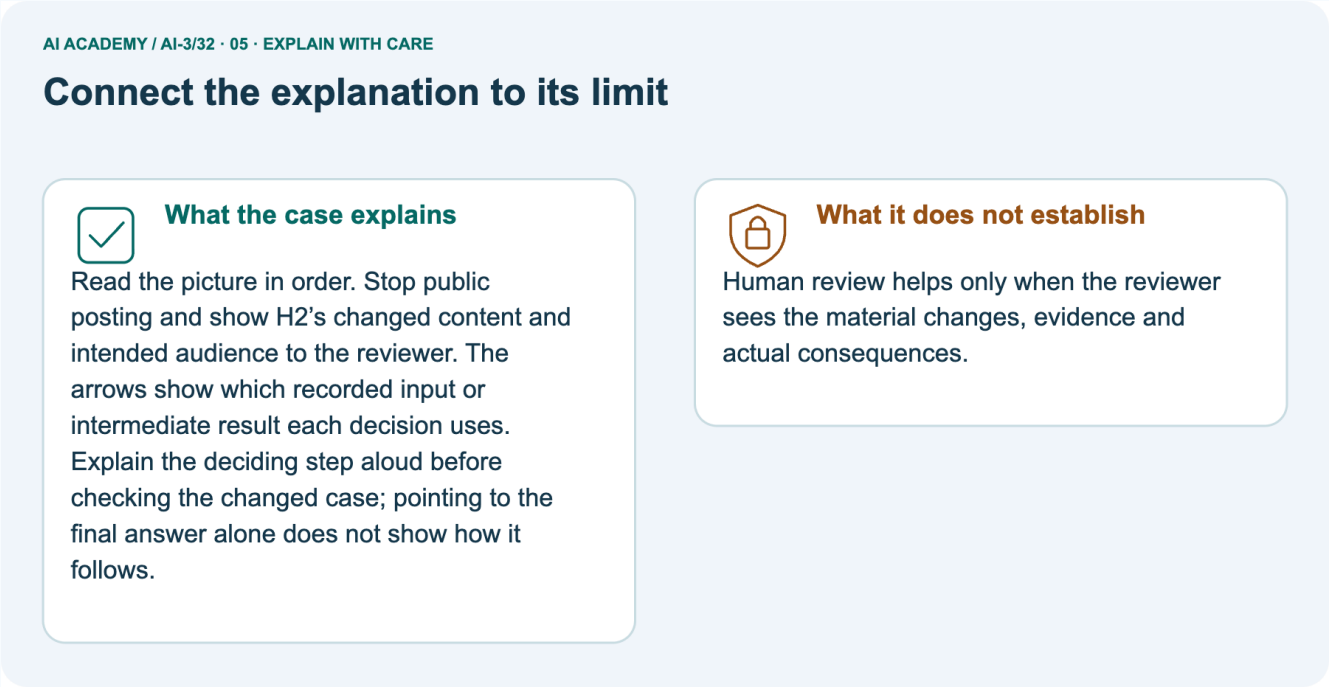
Expected result: The original approval covers that declared action. Reason: Both the approved content and its distribution scope now match.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Both the approved content and its distribution scope now match.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Stop public posting and show H2's changed content and intended audience to the reviewer. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

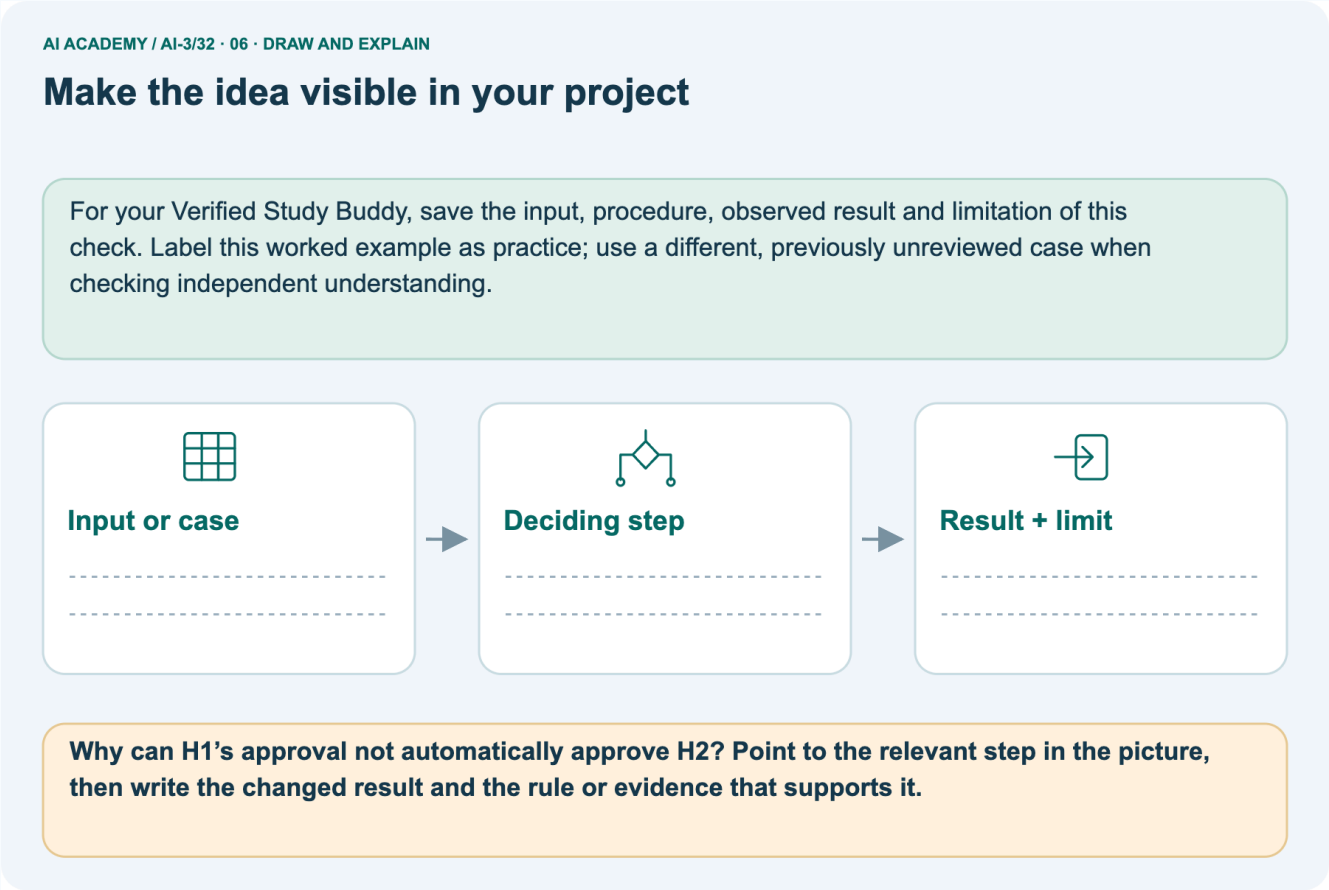
Limit: Human review helps only when the reviewer sees the material changes, **evidence** and actual consequences.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Human review helps only when the reviewer sees the material changes, **evidence** and actual consequences.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can H1's approval not automatically approve H2? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The reviewer did not approve the added personal detail or public audience. Approval must match the version and action. Changed-case check: The original approval covers that declared action.

Both the approved content and its distribution scope now match.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Human AI Workflow: the case

02. Follow the deciding steps

03. Human review and consequential use — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.